

A Proof of Conjecture 00000000047: Every $[n]$ Has a Prime Permutation

Feiyu*

Claude (Opus 5.5)[†]

2026-10-02

Abstract

Conjecture 00000000047 asserts that for every sufficiently large even n there is a permutation π of $[n] = \{1, \dots, n\}$ such that $\pi(i) + i$ is prime for all i . We prove that such a permutation exists for every n , even or odd, and that it can be chosen to be an involution. The proof is a short induction on Bertrand's postulate, and it is formalized in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000000047 reads:

Conjecture: For every sufficiently large even n there exists a permutation π of $[n]$ such that $\pi(i) + i$ is prime for all i (a “prime permutation”).

2 Reading of the statement

As usual, $[n] = \{1, 2, \dots, n\}$. A *prime permutation* of $[n]$ is a bijection $\pi: [n] \rightarrow [n]$ such that $\pi(i) + i$ is a prime number for every $i \in [n]$. The conjecture asserts that there is an N such that every even $n \geq N$ admits a prime permutation of $[n]$. There is no ambiguity to resolve.

3 Result

Theorem 1. *For every integer $n \geq 0$ there is an involution π of $[n]$ such that $\pi(i) + i$ is prime for all $i \in [n]$. In particular Conjecture 00000000047 holds, with any threshold N , and the evenness hypothesis is not needed.*

4 Proof

We use Bertrand's postulate in the form: for every $n \geq 1$ there is a prime p with $n < p \leq 2n$.

Proof of Theorem 1. By strong induction on n . For $n = 0$ the empty permutation works.

Let $n \geq 1$, and assume the statement for all smaller values. Choose a prime p with $n < p \leq 2n$ and put $m = p - n$, so that $1 \leq m \leq n$. Split

$$[n] = \{1, \dots, m-1\} \cup \{m, \dots, n\}.$$

*Independent researcher.

[†]Anthropic.

On the second block define $\pi(i) = p - i$. If $m \leq i \leq n$ then $m = p - n \leq p - i \leq p - m = n$, so π maps the block $\{m, \dots, n\}$ into itself; clearly $\pi(\pi(i)) = i$, and $\pi(i) + i = p$ is prime. The first block is $[m - 1]$, and $m - 1 = p - n - 1 < n$ because $p \leq 2n$. By the induction hypothesis there is an involution of $[m - 1]$ with the required property, and we let π act on the first block by it. The two pieces together form an involution of $[n]$, and $\pi(i) + i$ is prime for every $i \in [n]$. $\square$

Example 2. For $n = 10$ take $p = 11$, so $m = 1$ and $\pi(i) = 11 - i$ on all of $[10]$: the pairs $\{1, 10\}, \{2, 9\}, \{3, 8\}, \{4, 7\}, \{5, 6\}$ all sum to 11. For $n = 8$ take $p = 11$, so $m = 3$: the block $\{3, \dots, 8\}$ is paired as $\{3, 8\}, \{4, 7\}, \{5, 6\}$, and the remaining $[2]$ is handled with $p = 3$, giving the pair $\{1, 2\}$. So $\pi = (1\ 2)(3\ 8)(4\ 7)(5\ 6)$ works for $n = 8$.

5 Formalization

The accompanying Lean 4 project (`lean/Submission/Basic.lean`) states the conjecture and proves it against Mathlib.

- `HasPrimePermutation n` — there is $\pi : \text{Equiv.Perm (Finset.Icc 1 n)}$ with $\text{Nat.Prime } (\pi\ i + i)$ for every i . A permutation of the finite set $\{1, \dots, n\}$, as a type, is exactly a permutation of $[n]$.
- `ConjectureHolds` — for some N , every even $n \geq N$ satisfies `HasPrimePermutation n`.
- `exists_prime_involution` — the induction above, producing a function $f : \mathbb{N} \rightarrow \mathbb{N}$ that maps $[n]$ into itself, satisfies $f(f(i)) = i$ and makes $f(i) + i$ prime for $i \in [n]$. It uses Mathlib’s Bertrand postulate `Nat.exists_prime_lt_and_le_two_mul`.
- `hasPrimePermutation` — every n admits a prime permutation, obtained from f via `Function.Involutive.toPerm`.
- `conjecture_00000000047` — `ConjectureHolds`, with threshold $N = 0$.

Primality is Mathlib’s `Nat.Prime`. The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms:

```
#print axioms conjecture_00000000047
```

reports exactly `propext`, `Classical.choice` and `Quot.sound`.

6 Remarks

Remark 3 (fixed points). For even n the involution constructed above has no fixed points, so it splits $[n]$ into $n/2$ pairs with prime sums. Indeed a fixed point needs $p = 2i$, so $p = 2$, which happens only in the case $n = 1$. When n is even, the paired block $\{m, \dots, n\}$ has $2n - p + 1$ elements, an even number because p is odd. So the leftover block $[m - 1]$ again has an even number of elements, and the case $n = 1$ never arises. This pairing statement for even n is a classical consequence of Bertrand’s postulate; see Greenfield and Greenfield [1], who also discuss harder variants. For odd n a fixed point is unavoidable for an involution, and our construction places it at $i = 1$, where $1 + 1 = 2$ is prime. This fixed-point refinement is not part of the Lean development.

Remark 4. The conjecture asks only for large even n , which suggests a probabilistic argument was intended. Nothing of the kind is needed: the deterministic construction covers all n . A related but genuinely different question is whether the prime-sum graph on $[n]$ has a *Hamilton cycle* (Conjecture 00000000045). There the vertices must form a single cycle, not a union of transpositions, and parity already rules it out for odd n .

References

- [1] L. E. Greenfield and S. J. Greenfield, *Some problems of combinatorial number theory related to Bertrand's postulate*, J. Integer Sequences 1 (1998).